



Course Title: Technology, Environment and Society

HUM 4147

Content (s)

- Climate and Climate Change
- Global Warming and Green House Effect
- Depletion of Ozone Layer and it's Impact
- Acid Precipitation

Climate and Climate Change





Climate and Climate Change

Atmosphere – the envelope of gases surrounding the earth or another planet.

Weather – what is happening right now in the atmosphere. It's a combination of Temperature, Humidity, Precipitation, Cloudiness, Visibility, Wind and Atmospheric pressure.

Or

Weather is the condition of the outside air or atmosphere at a particular time and place

Climate – is the statistics of the environment over time. It can include seasonal cycle, extreme events, year to year variations

Overall weather pattern of precipitation and air temperature in a specific area over time.



Climate and Climate Change

Climate change refers to any significant change in measures of climate (such as temperature, precipitation, or wind) lasting for an extended period (decades or longer).

- Climate change may result from:
 - natural factors, such as changes in the sun's intensity or slow changes in the Earth's orbit around the sun;
 - natural processes within the climate system (e.g. changes in ocean circulation);
 - **human activities** that change the atmosphere's composition (e.g. through burning fossil fuels) and the land surface (e.g. deforestation, reforestation, urbanization, desertification, etc.)



Climate and Climate Change

Seven Theories of Climate Change

Anthropocentric Global Warning Theory: These theory of climate change contends that human emissions of green house gases, principally carbon dioxide (CO₂, methane, and nitrous oxide, are causing a catastrophic rise in global temperatures. The mechanism whereby this happens is called the enhanced **greenhouse effect**. Includes the activities of human beings. For example: Usage of Non renewable energy resources, Usage of artificial fertilizers and insecticides, General environmental pollution

2. Bio-thermostat Theory

Carbon Sequestration

The more CO₂ there is in the air, the better plants grow and the more CO₂ they remove from the air and store in their leaves, branches, trunks, and roots, as well as in the soil beneath the plants – a suite of processes called “**sequestration**.” Works for natural balance.

Carbonyl Sulfide

Carbonyl sulfide (COS) is a biologically produced sulfur gas emitted from soils.

Diffuse Light

higher levels of CO₂ promote greater plant productivity, plants emit greater amounts of gases converted into aerosols called “biosols.”



Climate and Climate Change

3. Cloud Formation and Albedo

The albedo refers to the reflectivity of a surface such as the snow, ice, sand or the clouds which reflect up to 80 percent of the solar radiation back into the space. It postulates that changes in the formation and albedo of clouds create negative feedbacks that cancel out all or nearly all of the warming effect of higher levels of CO₂.

4. Human Forcing Besides Greenhouse Gases

transformation of Earth's surface by clearing forests, irrigating deserts, and building cities.

Urban Heat Islands

Cities tend to be warmer than suburbs, and suburbs warmer than rural areas, because they have greater concentrations of energy-producing machines and vehicles and large amounts of concrete, asphalt, and other building and road materials that absorb solar energy and then re-emit thermal energy.

Aerosols and ozone

Anthropogenic aerosols and ozone have shorter lifetimes than greenhouse gases, and therefore their concentrations are higher in source regions and downwind.



Climate and Climate Change

Deforestation

releases CO₂ into the atmosphere and prevents forests from sequestering carbon in the future. The pasture or crop land that replaces the forest lacks the shade created by a forest canopy and tends to be warmer.

Coastal development

Anthropogenic activities in coastal areas such as logging, agriculture, construction, mining, drilling, dredging, and tourism all can increase or (more rarely) decrease surface temperatures of nearby bodies of water.

Jet contrails

jets often leave trails behind them, called contrails (short for “condensation trails”). Composed of water vapor, they precipitate the creation of low clouds that have a net warming effect.

5. Ocean Current

When the Thermohaline Circulation (THC) is relatively weak the earth-system typically has less net evaporation cooling and less deep ocean upwelling of cold water. At these times, energy accumulates in the ocean’s upper mixed layer and over a period of a decade or two the global ocean begins to warm.



Climate and Climate Change

6. Planetary Motion

How the Earth's tilt, orbit and other planets effect the climate, by natural gravitational and magnetic oscillations of the solar system induced by the planet's movement through space. These oscillations modulate solar variations and/or other extraterrestrial influences of Earth, which then drive climate change

The shape or "eccentricity" of Earth's orbit also varies on cycles of 100,000 and 400,000 years due to the tug of other planets, specifically Jupiter and Saturn, on Earth. **More "tilt" roughly means warmer northern hemisphere summers and colder winters; less "tilt" means cooler summers and milder winters.**

7. Solar Variability

How changes in the coronal ejections and magnetic fields of the sun cause changes in cloud formation, ocean currents, wind etc. cause climate change. **Changes in the brightness of the sun are caused by sunspots – bursts of energetic particles and radiation – that vary in frequency in cycles of roughly 11, 87, and 210 years. These cycles cause changes in the amount of electromagnetic radiation – also called "solar wind" – that reaches Earth and its atmosphere, which in turn affects Earth's climate** Changes in cloud formation are linked to variations in sea surface temperatures, wind patterns, and the oceans' thermohaline circulation.

Global Warming and Green House Effect





Global Warming and Green House Effect

Comparative Discussion



Mars

Thin atmosphere...little
heat to trap...Frozen planet



Venus

Thick atmosphere...much
heat to trap...Hottest planet



Earth

Just right atmosphere
...enough heat to trap for living

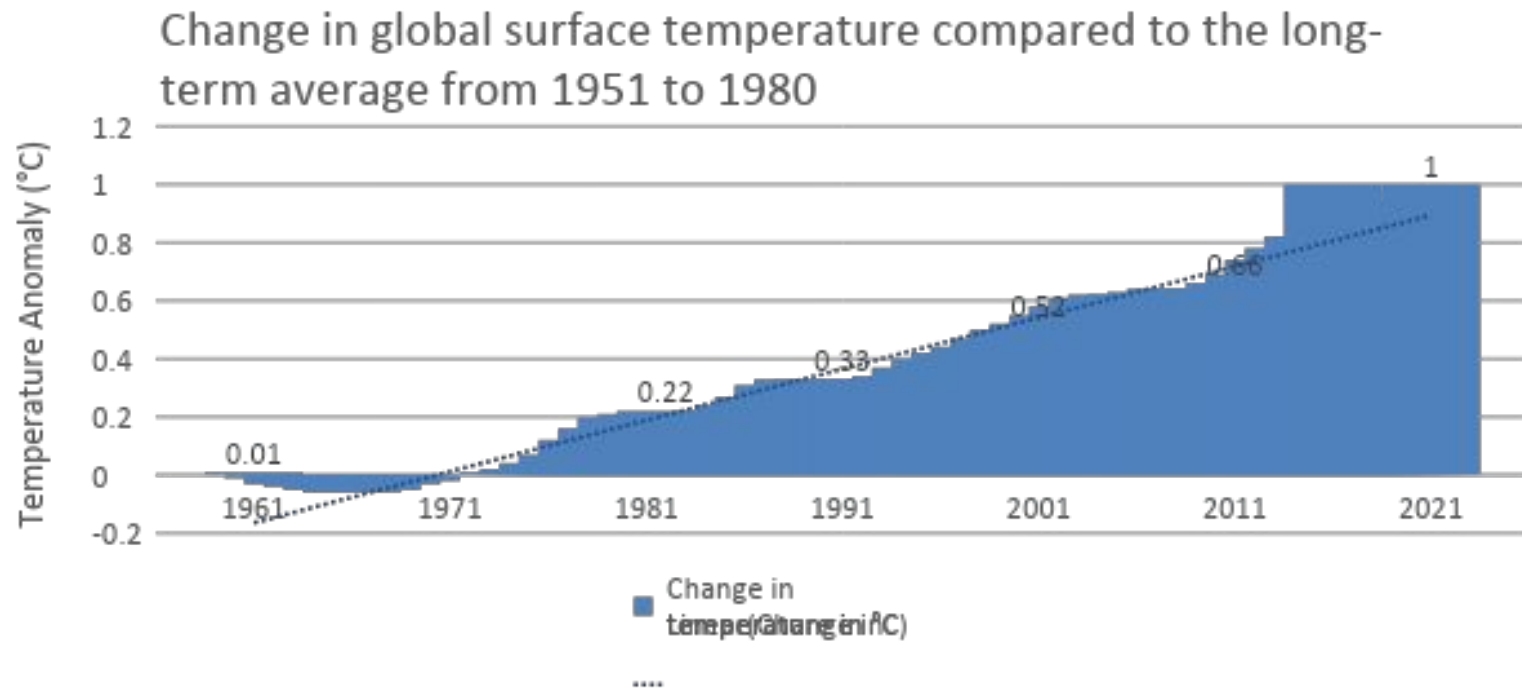


WHAT IS GLOBAL WARMING?

Global Warming is the **long-term heating of the Earth's surface** due to various human activities.

- Sometimes Global Warming and Climate Change are **used interchangeably but they are different**
- **Due to climate change**, various changes are occurring and one of them is that the **Earth's temperature is increasing**
- To put it into simple terms, Climate Change is the **cause** and Global Warming is the **result**

GLOBAL TEMPERATURE CHANGE



GREENHOUSE EFFECT AND CLIMATE CHANGE

GREENHOUSE EFFECT IS THE WAY IN WHICH HEAT IS TRAPPED CLOSE TO THE EARTH'S SURFACE BY GREENHOUSE GASES.

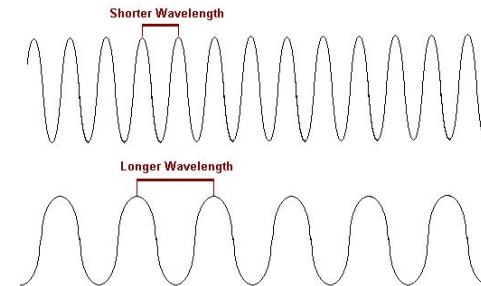
GHGS INCLUDE CO₂, CH₄, NO₂, WATER VAPOR ETC.



Global Warming and Green House Effect

Short wavelength – More energy

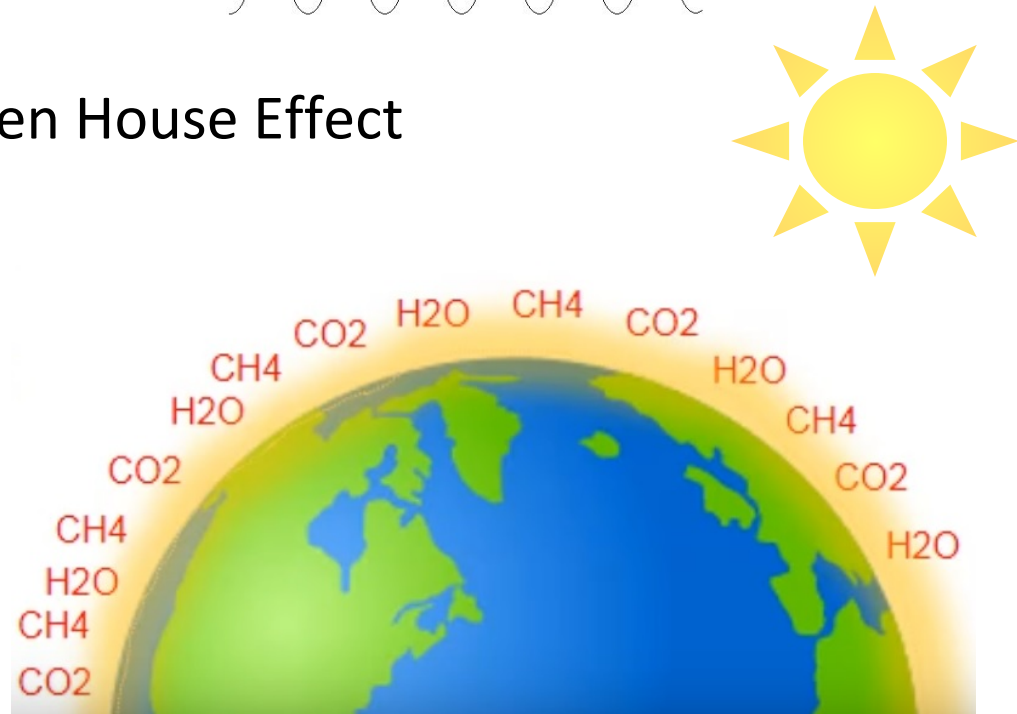
Long wavelength – less energy



Heat Trap Happens – Green House Effect

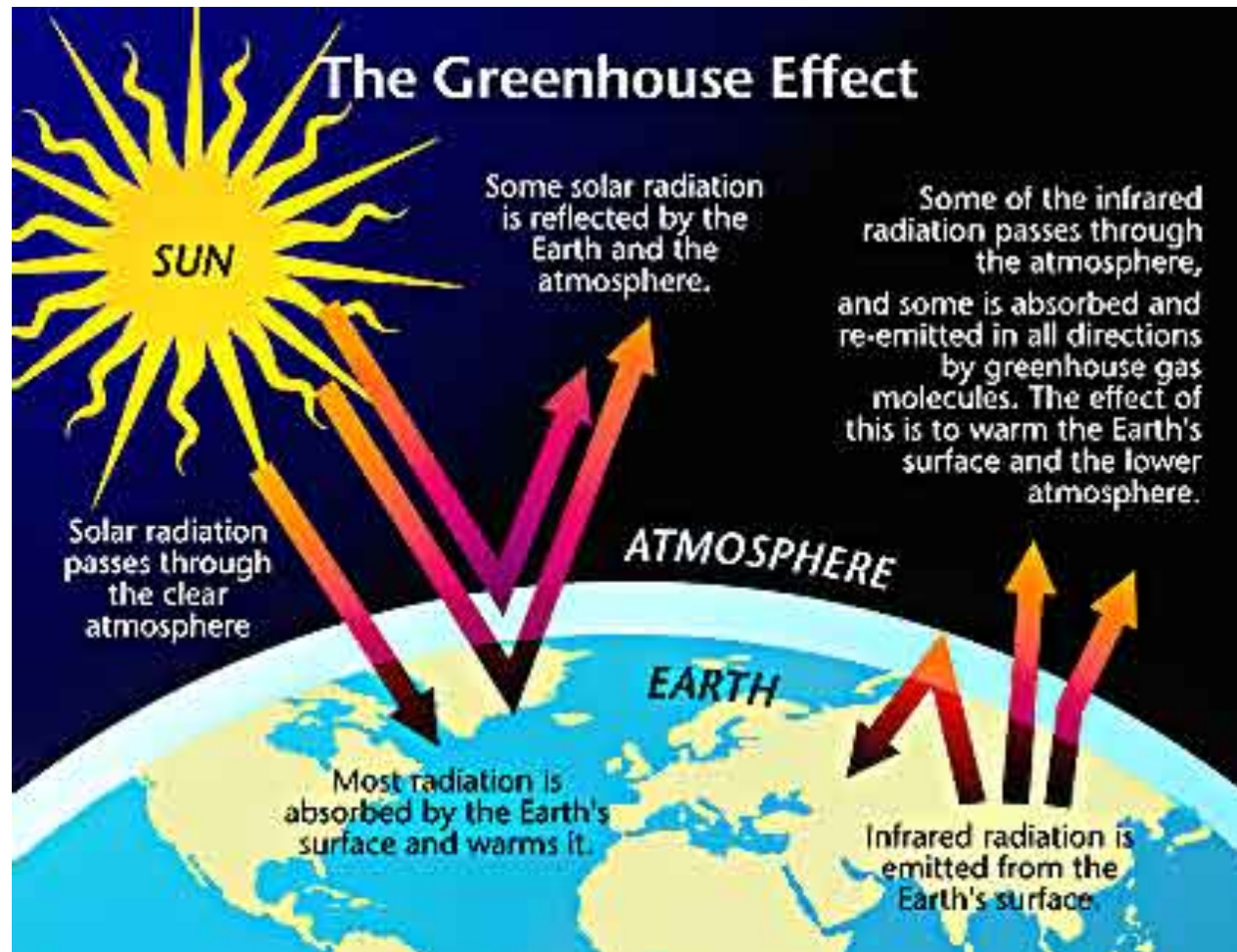
Factors influencing – **greenhouse cover**, color of the surfaces inside the greenhouse, and type of surfaces inside

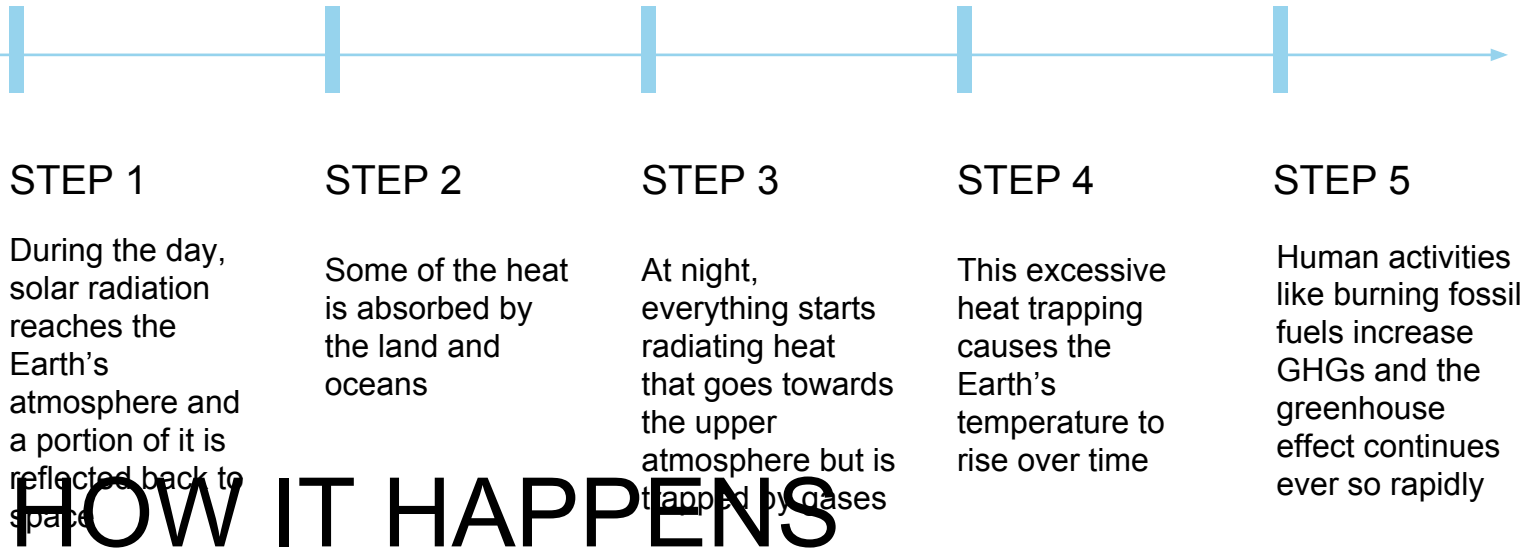
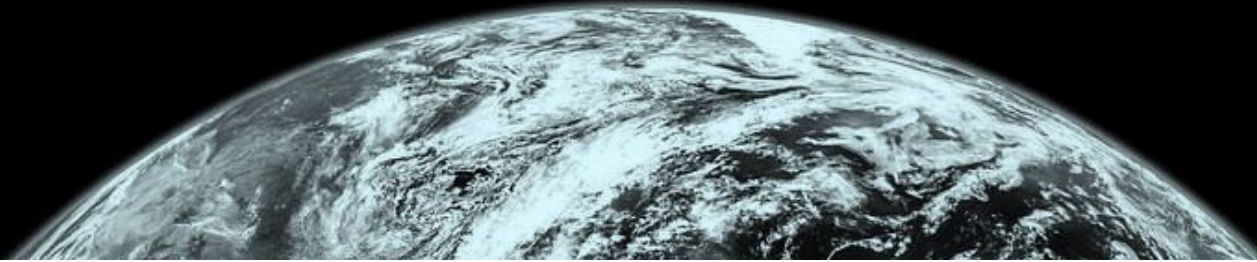
water vapor (H_2O), carbon dioxide (CO_2), methane (CH_4), and nitrous oxide (N_2O) and Ozone (O_3)



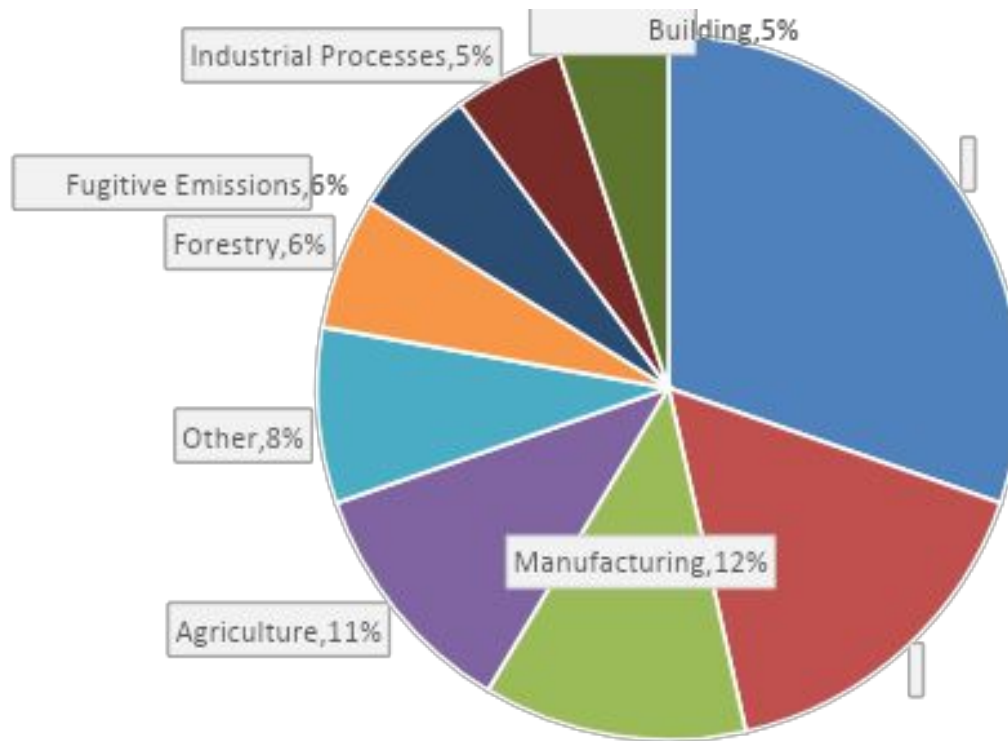


Global Warming and Green House Effect





GHG EMISSIONS BY SECTOR





Global Warming and Green House Effect

Where is the problem then???

✓ Earth is getting warmer

Human Actions (Anthropogenic)

- Burning of fossil fuels (CO_2)

 - Automobiles

 - Industry

 - Homes

- Agriculture

 - Overuse of fertilizers (NO_2)

 - Over production of farm animals (CH_4)

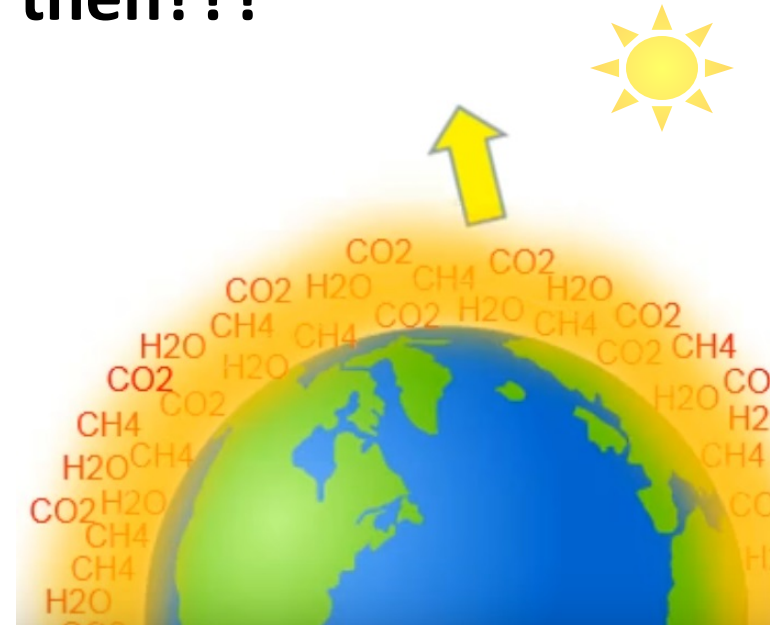
- Deforestation

 - Reduced photosynthetic plants that absorbs CO_2

- Water Pollution

 - Reduced photosynthetic algae absorbs CO_2

- Population Explosion



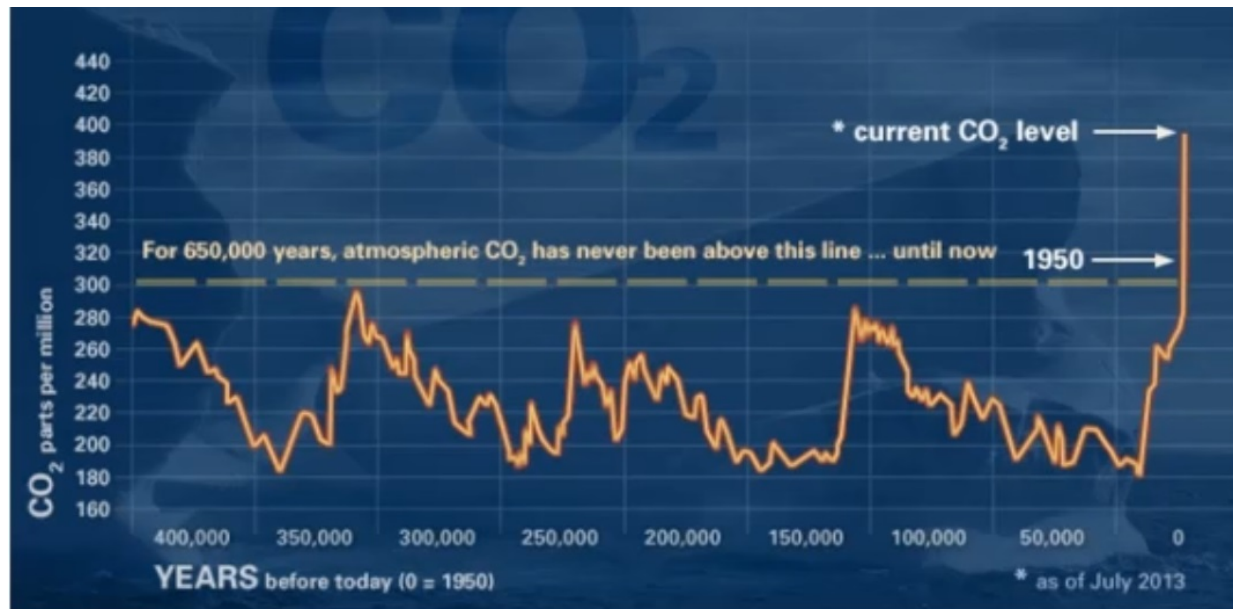


Global Warming and Green House Effect

Consequences

- ✓ Melting of glaciers and ice caps
- ✓ Rising sea level
- ✓ More severe weather and storms
- ✓ Droughts
- ✓ Reduction of crop production
- ✓ Spread of diseases
- ✓ Coral bleaching

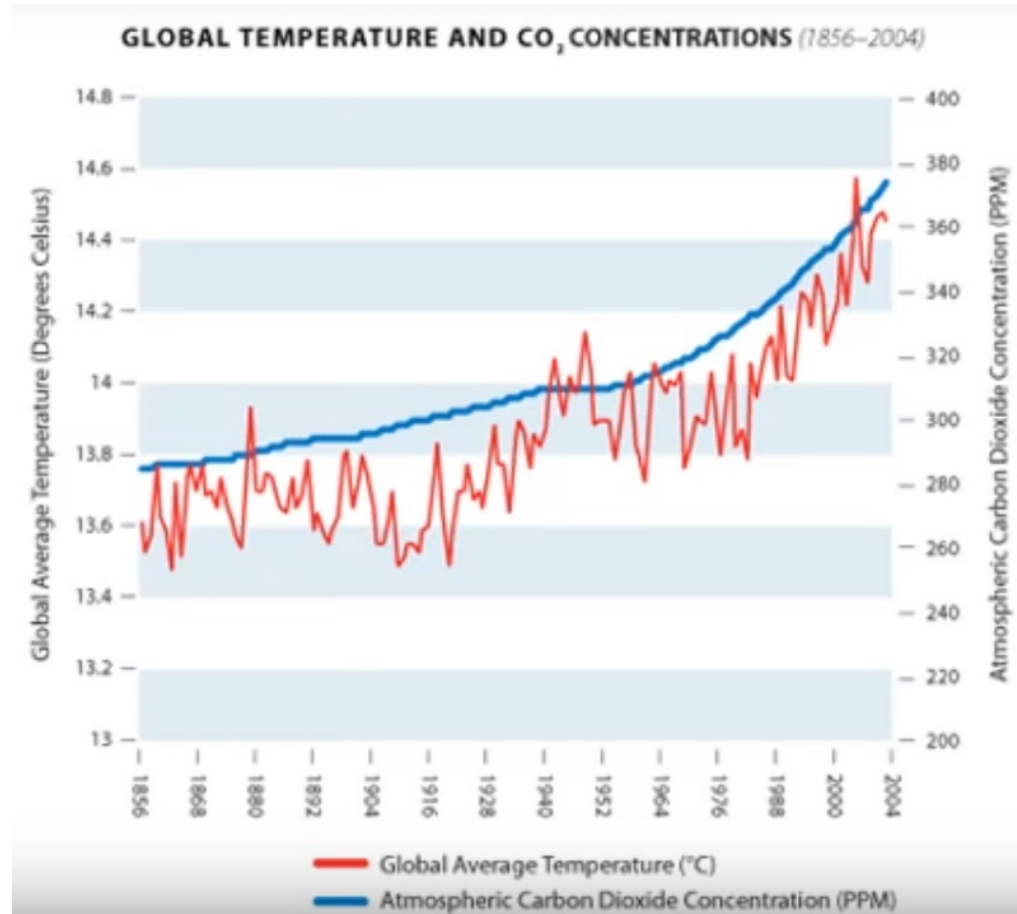
Since the **industrial revolution**, the rate CO_2 in the atmosphere has greatly increased





Global Warming and Green House Effect

Correlation between CO₂ increase and temperature hike; **Scientific evidence**





Global Warming and Green House Effect

Action to Take

Global Cooperation

Paris Agreement: Nations have pledged to reduce their green house gas emission

Reduce fossil fuel usage

better fuel efficiency

Bike use/walk

using public transport

adjust heat

alternative energy

Reforestation

Promotion of Organic farming

SO WHAT DO WE DO?

REDUCE, REUSE, RECYCLE



Production takes a lot of energy. By implementing 3Rs, we can reduce our carbon footprint.

SAVE ENERGY AT HOME



Turn things off when not in use and use energy saving equipment.

CHANGE TRANSPORT METHODS



Use public transport more instead of personal cars. Or just use a bicycle.

PLANT MORE TREES



Trees and forests must be saved at all costs. Plant more trees wherever possible.

EDUCATE PEOPLE AROUND US



Awareness is a must. People should be educated about climate change and its dangers.



AREAS IN WHICH WE SHOULD FOCUS MOST

REDUCING THE PRODUCTION OF GREENHOUSE GASES

Greenhouse gases are one of the **key reasons of climate change**. So, they are one of the first things that should be taken care of.

USING RENEWABLE ENERGY SOURCES

Producing energy using fossil fuels create a lot of problems. They make GHGs and there's also the fact that **we will run out of them soon**. Usage of renewable energy everywhere should be made a priority.

Ozone Depletion





Ozone Depletion

What is the “ozone layer?”

How does it protect us?

How did it come about?

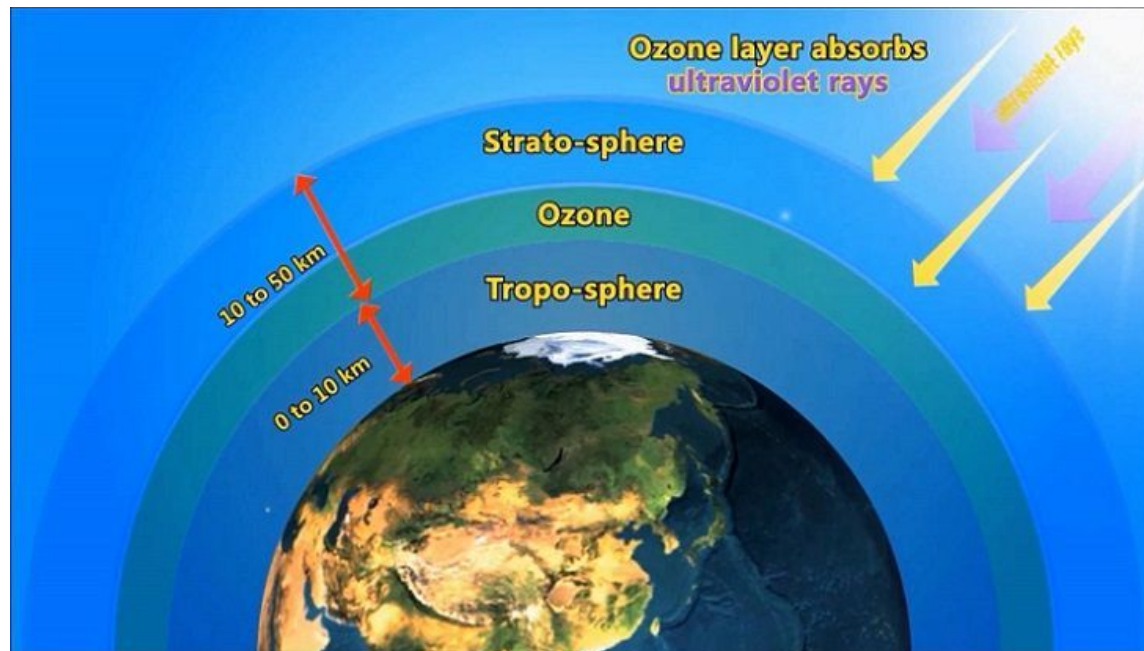
Recommended Video to watch

<https://www.youtube.com/watch?v=nXndl6cqWEw>



Depletion of Ozone Layer

- Found exists 9.3 to 18.6 miles (15 to 30 kilometers) above Earth, and serves as a shield from the harmful ultraviolet B (UVB) radiation emitted by the sun





Ozone Layer

- identified in 1839 by Christian Schonbein, a Swiss chemist
- named it ozone, from the Greek term “to smell”
- French scientists Charles Farby and Henri Buisson are credited with actual discovery of the Ozone Layer in 1913

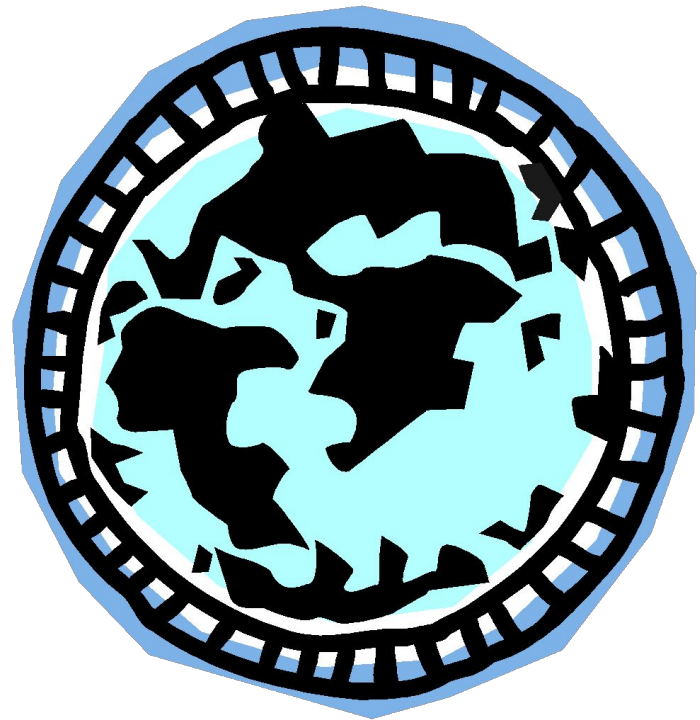


Depletion of Ozone Layer

Evolution of the Ozone Layer

Early planet history:

- no ozone present
- UV light directly hit planet's surface
- Oceans provided only refuge from UV radiation



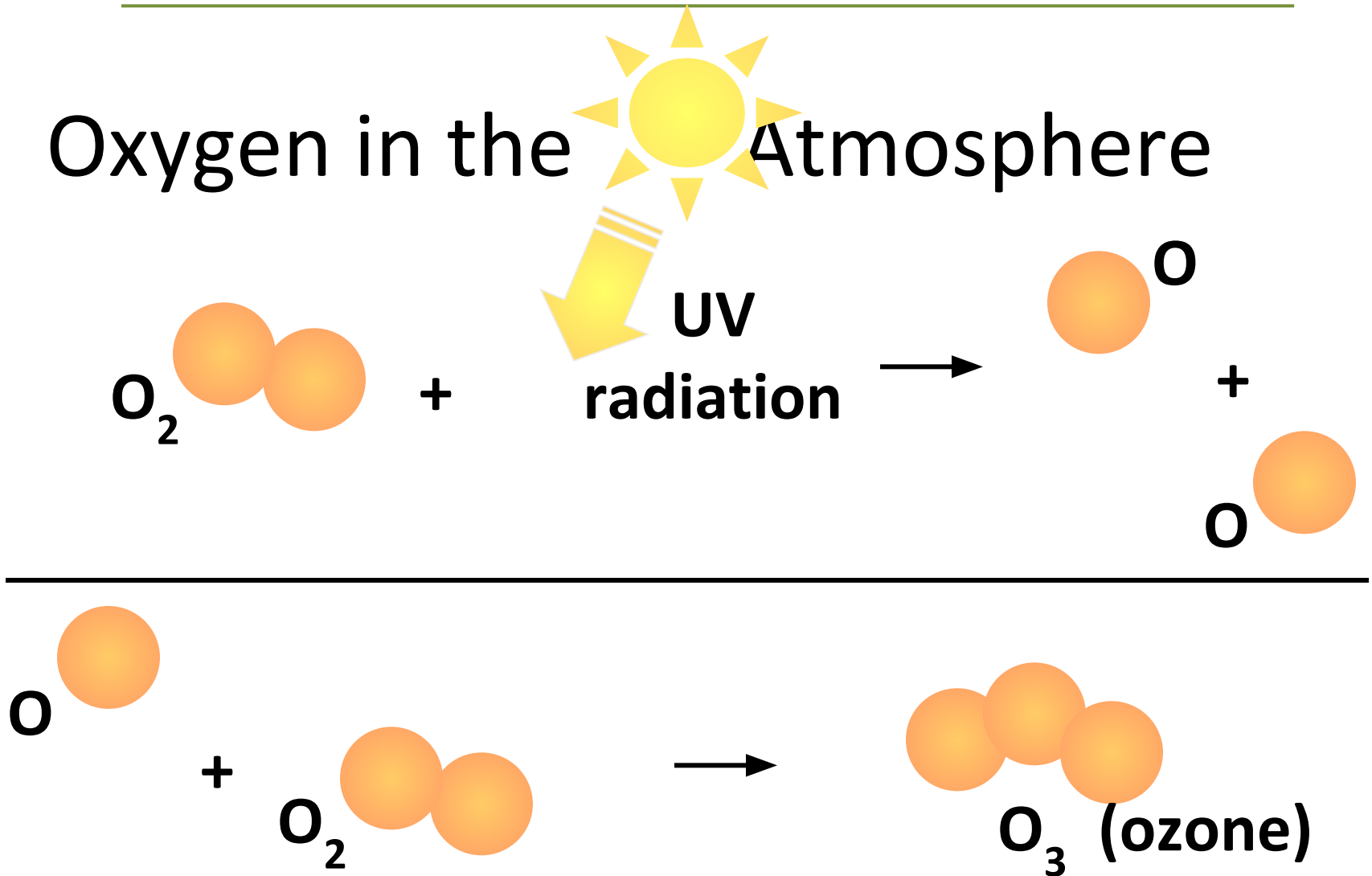


Depletion of Ozone Layer

Formation and Breakdown of Ozone

- made up of three oxygen atoms bonded together
- Ozone is a pale blue gas irritating to the nose and throat
- it is explosive and toxic. It is produced and destroyed naturally in the atmosphere and until recently
- this resulted in a nicely balanced equilibrium
- It is **formed** when oxygen molecules absorb ultraviolet radiation with **wavelengths less than 240 nanometers** and is **destroyed** when it absorbs ultraviolet radiation with **wavelengths greater than 290 nanometres.**

Oxygen in the Atmosphere



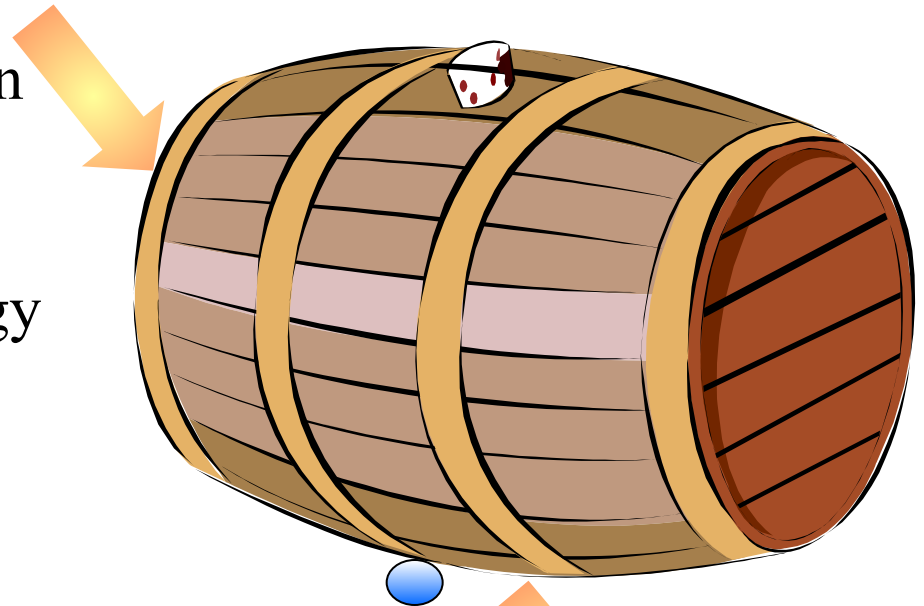


Depletion of Ozone Layer

Dynamic Equilibrium

- Free radicals with oxygen molecules to ozone
- Absorption of light energy radiation also triggers decomposition of ozone

creation of ozone

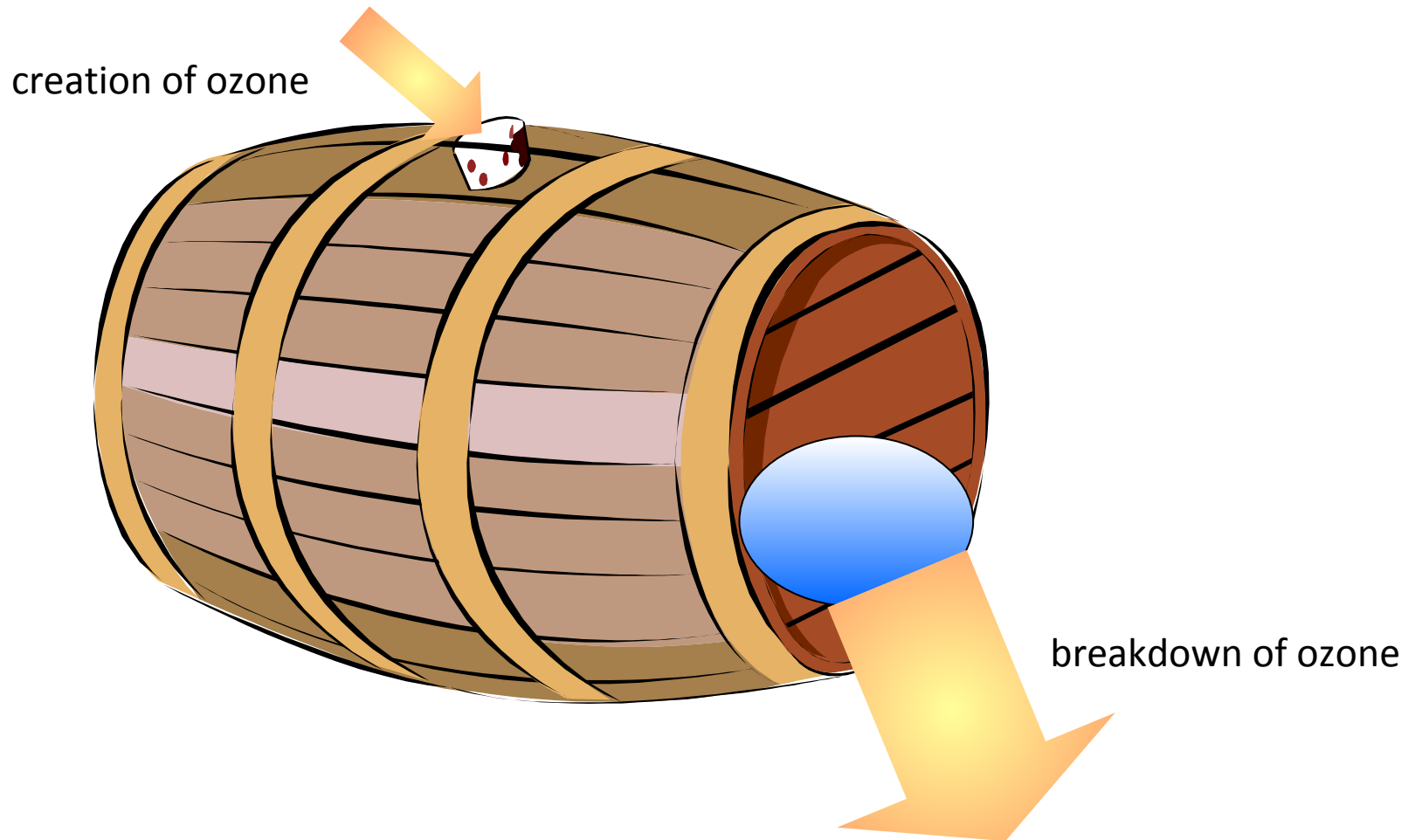


breakdown of ozone



Depletion of Ozone Layer

Anthropogenic Ozone Depletion





Modern Impacts to Ozone

Chlorofluorocarbons (CFCs)

- What are they?
- How do they impact the ozone layer?



Depletion of Ozone Layer

Development of CFCs

1928: DuPont scientists develop CFCs

“ideal compounds” for

refrigerants and propellants



WHY??



Chlorofluorocarbons (CFCs)

- refrigerators, automobiles and air-conditioners
- Cleaning agents for semi-conductors and precision parts
- Foaming agents for insulating materials and packing cushions
- Propellants for aerosol sprays





Depletion of Ozone Layer

CFCs as Refrigerants

Traditional Refrigerants vs.

(ammonia, sulfur dioxide, methyl chloride)

- Highly volatile
- Caustic and toxic
- Remove heat through vaporization of liquefied gas (only adequate as refrigerants)
- Expensive
- Heavy (transport, storage)

CFCs

- Non-flammable
- Non-toxic
- Trap heat (good insulators!)
- Inexpensive
- Light
- Extremely stable, inert



Depletion of Ozone Layer

CFCs as Propellants

- Light weight
 - Extremely stable or “inert”
-



What are the consequences of these two physical characteristics?

- CFCs likely to migrate upwards
- Too light to precipitate out with rainfall
- 5-15 years to migrate to stratosphere

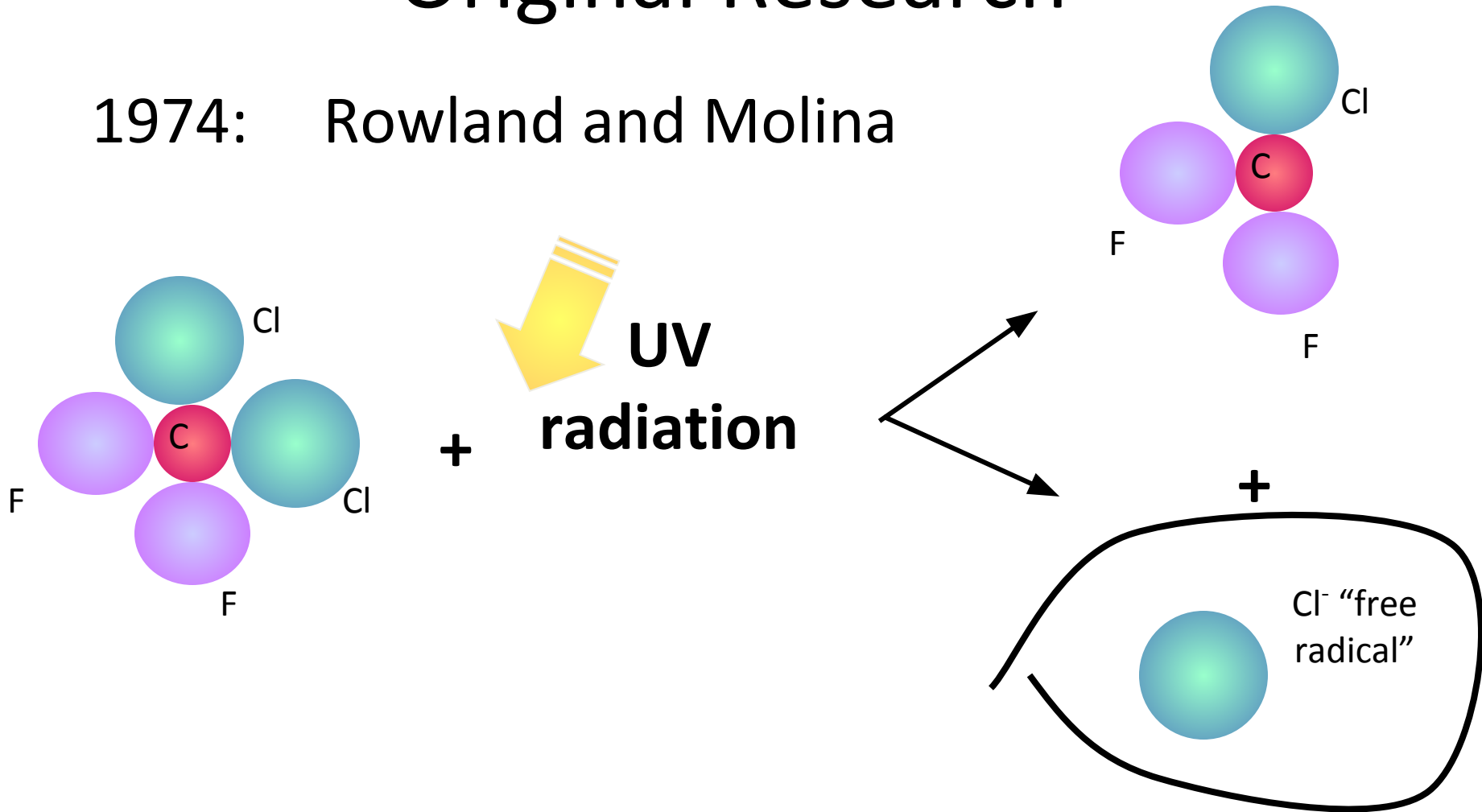


Marketing of CFCs

- 1958: DuPont releases CFCs on the market commercially
- 1971: James Lovelock speculates that CFCs put into the atmosphere may still be present
- 1973: Mario Molina and F. Sherry Roland start to investigate

Original Research

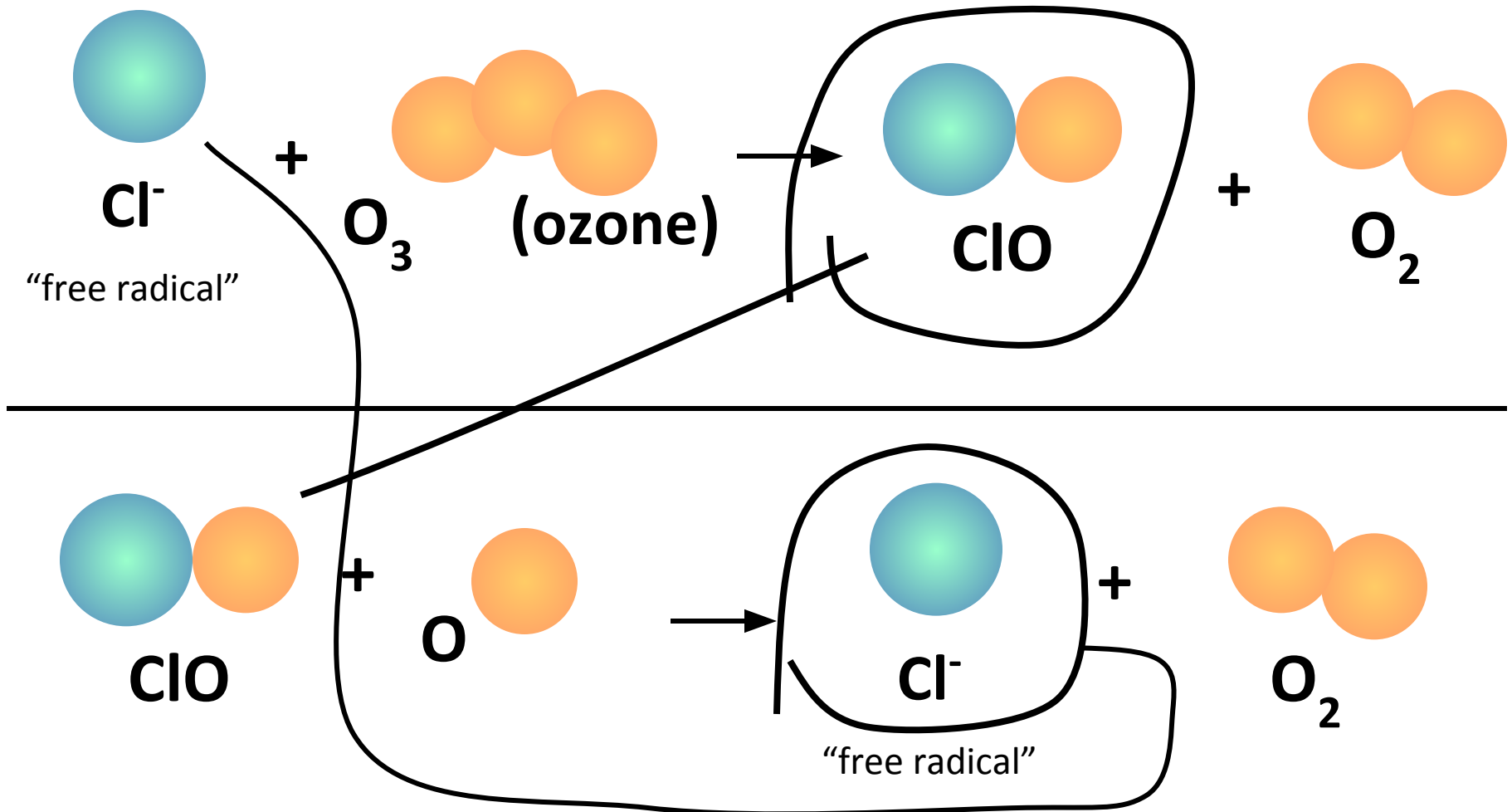
1974: Rowland and Molina





Depletion of Ozone Layer

Cl⁻ "Free Radicals" ...





Depletion of Ozone Layer

In the news...

1974: Molina and Rowland publish their hypothesis in Nature.

New York Times runs front page

DuPont responds with study showing that CFCs in *troposphere* are benign



Depletion of Ozone Layer

High Risk and Political Savvy

1975: 200% increase in CFC use from
1968, only eight years

1979: The FDA, EPA of USA ban non-essential
uses of CFCs !

First time substance EVER banned
without direct proof of harm

1982: 20 other countries join US in ban of CFCs



Scientific Controversies

1982: British science teams in Antarctica observe 20% decline in O₃ layer

US scientists relying on TOMS	(Total
Ozone Mapping	Spectrometer)
measurements from	space claim to
observe nothing	



Depletion of Ozone Layer

Scientific Evidence

1983: British scientists observe 30% reduction in ozone layer.

US scientists claims no reduction.

1985: British observe 50% reduction.

US claims no reduction.

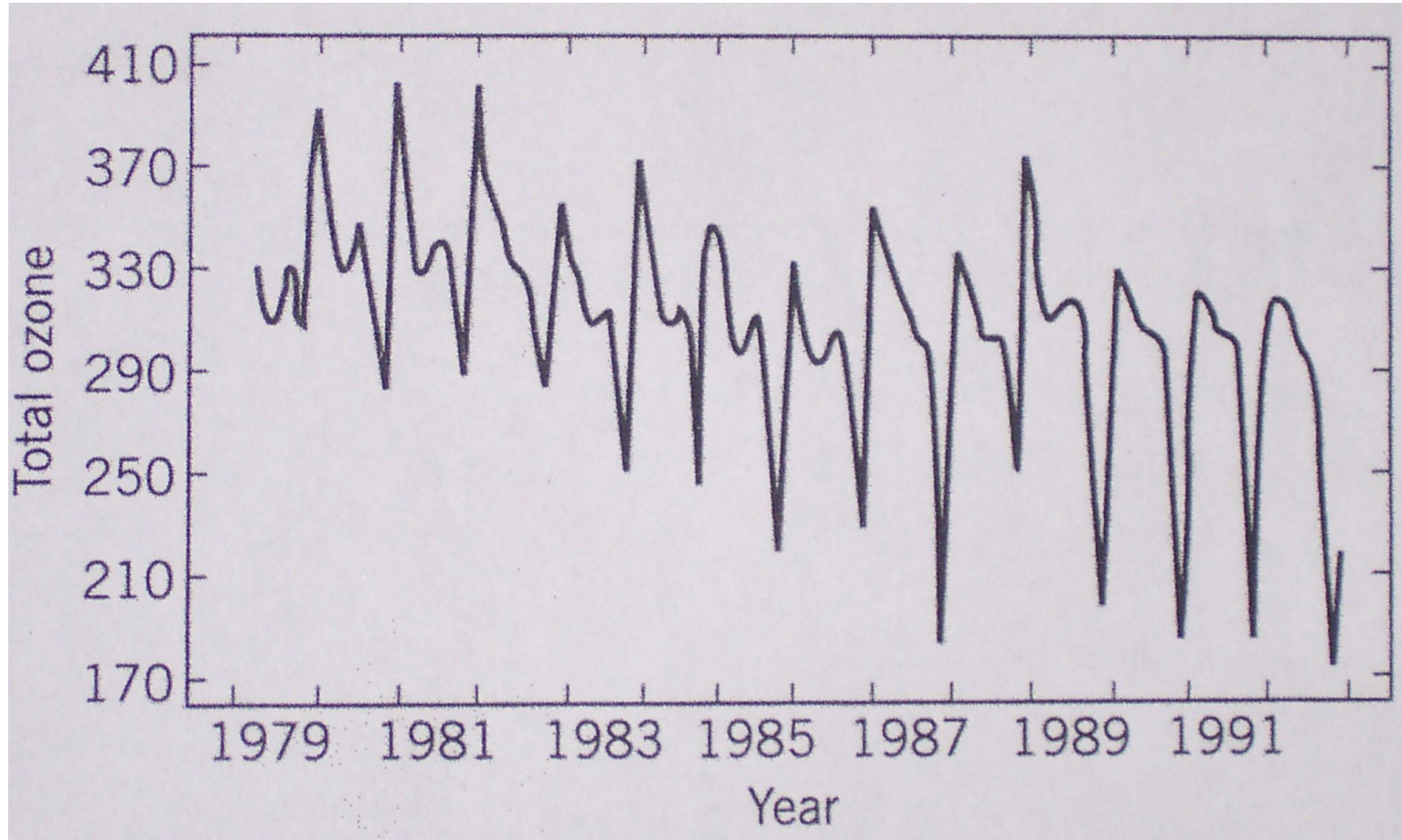
US re-tests and confirms.

WHY THE SCIENTIFIC SNAFUS??



Depletion of Ozone Layer

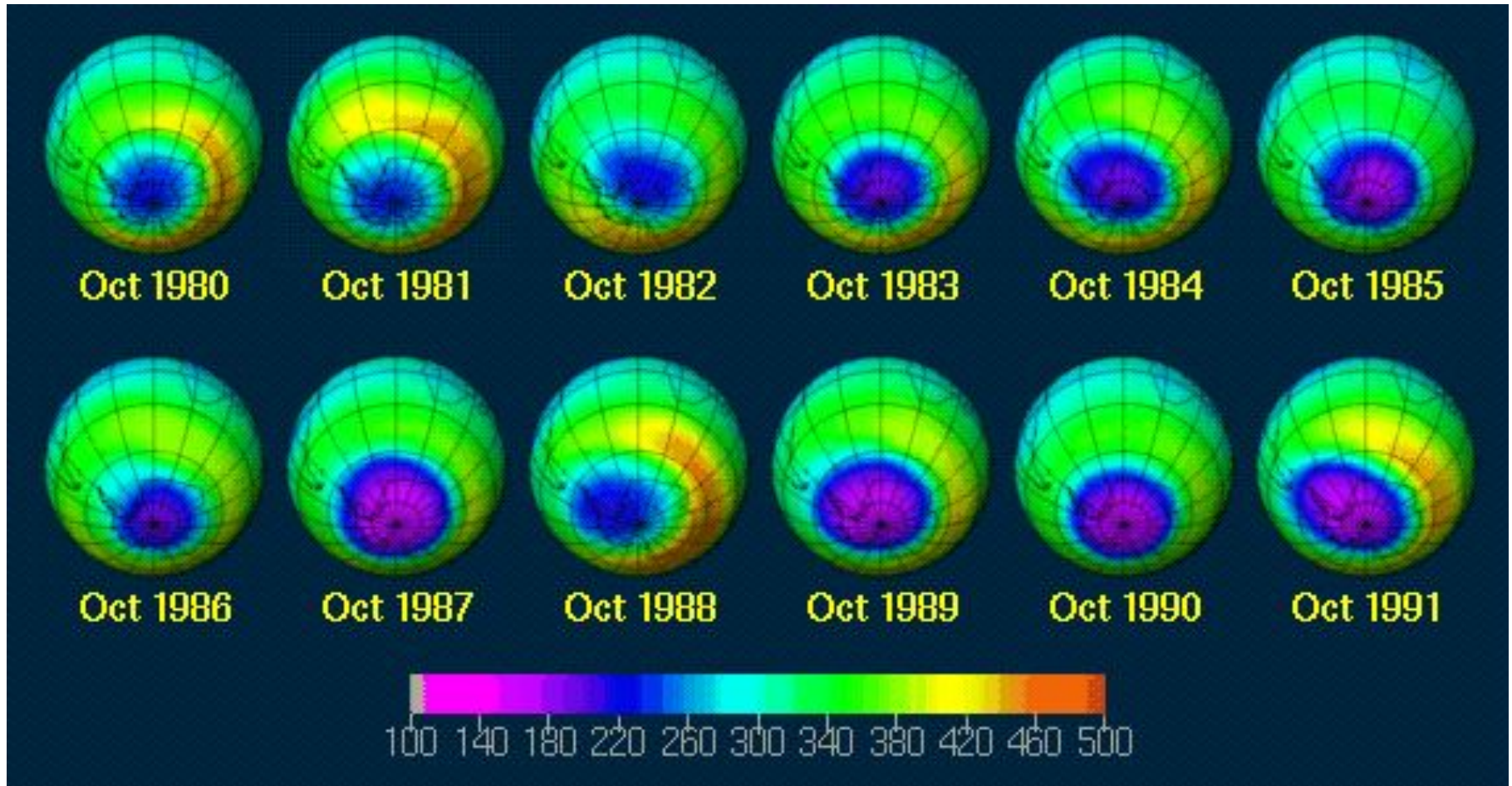
Total ozone





Depletion of Ozone Layer

TOMS Data (corrected)

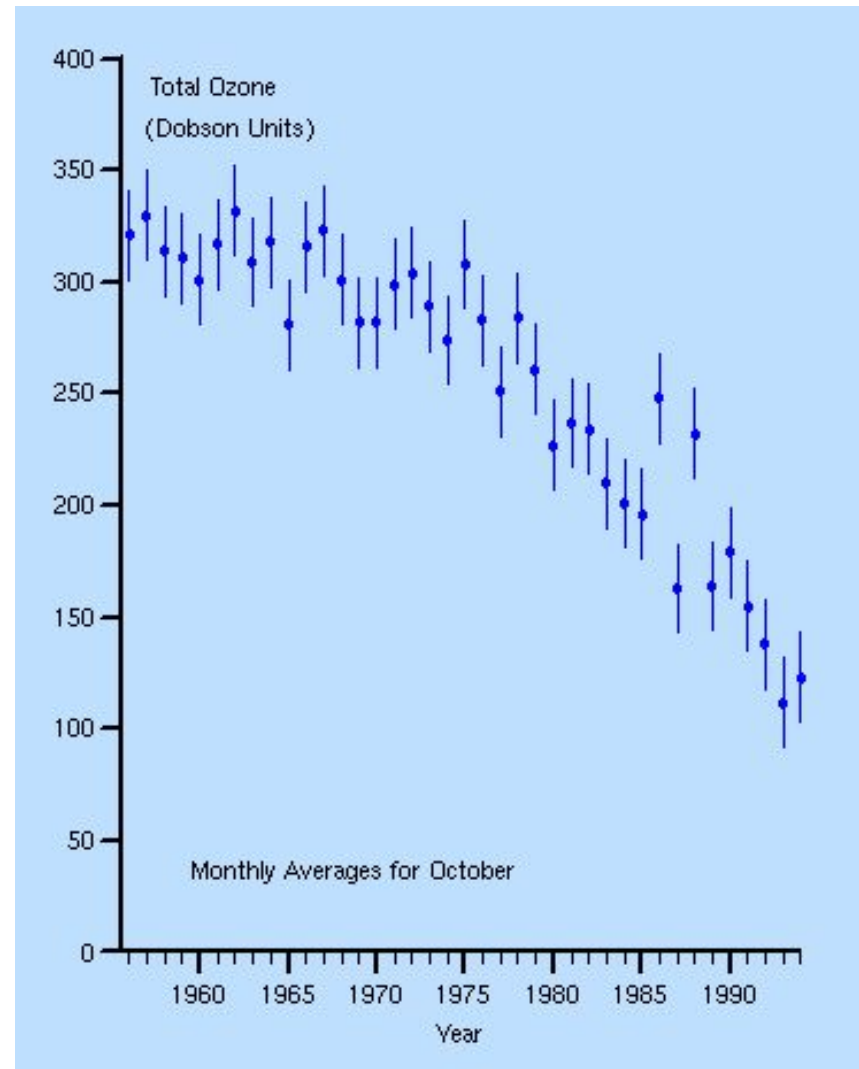




Depletion of Ozone Layer

October Average for Total
Ozone over Antarctica,
1955-1995

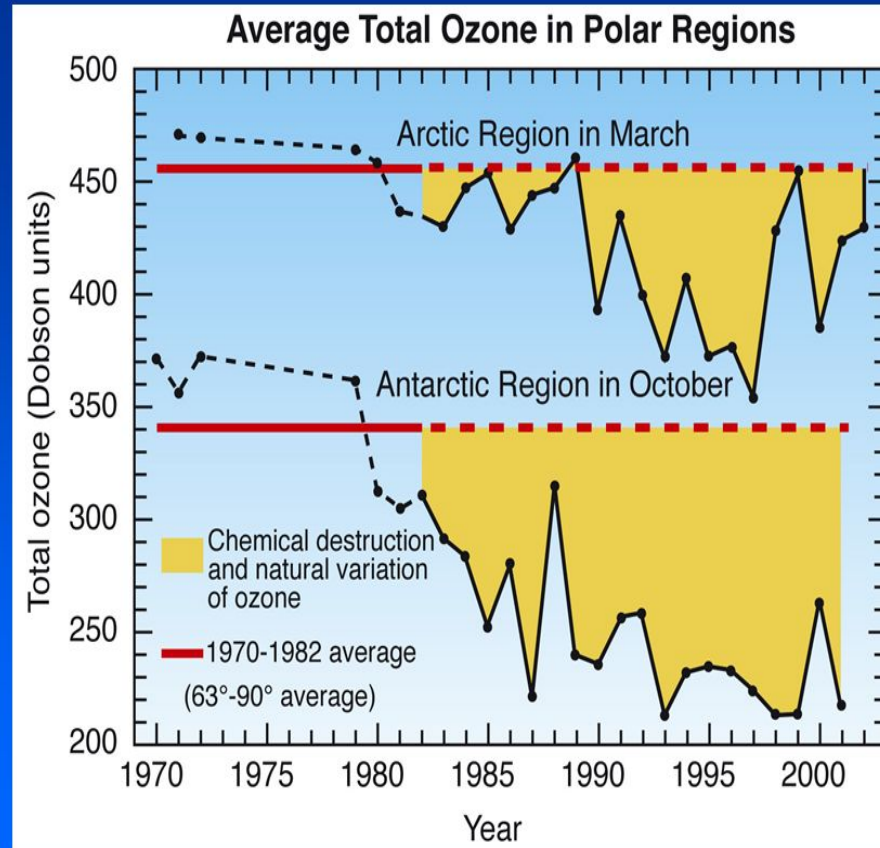
Based on British measurements
from weather balloons





Depletion of Ozone Layer

Average for Total Ozone over
Antarctica, 1995-2000





Montreal Protocol Landmark

1987: 2 yrs of intensive research reveal that **ozone hole is anthropogenic**

1988: UN hold meeting in Montreal
45 Nations sign to reduce CFC use
by 50% by year 2000.

Developing countries' efforts
would be 'subsidized'



Depletion of Ozone Layer

Two steps forward...

1990- Follow up meetings result in:

1992: Industrialized nations: total ban by 2000

Developing nations: ban by 2010, with assistance from developed nations

US agrees to complete phaseout by 1996;
DuPont to halt production by 1997

1995: Rowland and Molina receive Nobel Prize



Depletion of Ozone Layer

Spatial and Temporal Patterns

- Poles have greater ozone loss than other regions:
 - Colder
 - More vapor formation
 - Also: polar vortex
- Particularly severe in polar spring (October)
- Increased hydrogen would enhance this phenomenon



Depletion of Ozone Layer

Impacts of Ozone Depletion

Human Health

- Skin cancer
- Melanoma
- Cataracts
- Immune system function
- Increased incidence, severity and duration of infectious diseases
- Reduced efficacy of vaccinations

Ecological Health

- Pathogen locally up & down
- Biodiversity locally up & down
- Aquatic organisms adversely impacted
- Decreased biomass productivity
- Polar systems especially vulnerable



Impacts of Ozone Depletion

Economic

- Plastics
 - designed with stabilizers to withstand UV radiation of certain intensity
 - replacement of key medical equipment and supplies, decreased lifespan of plastics
- Manufacturing practices
- Agriculture
- Consumer costs and burdens

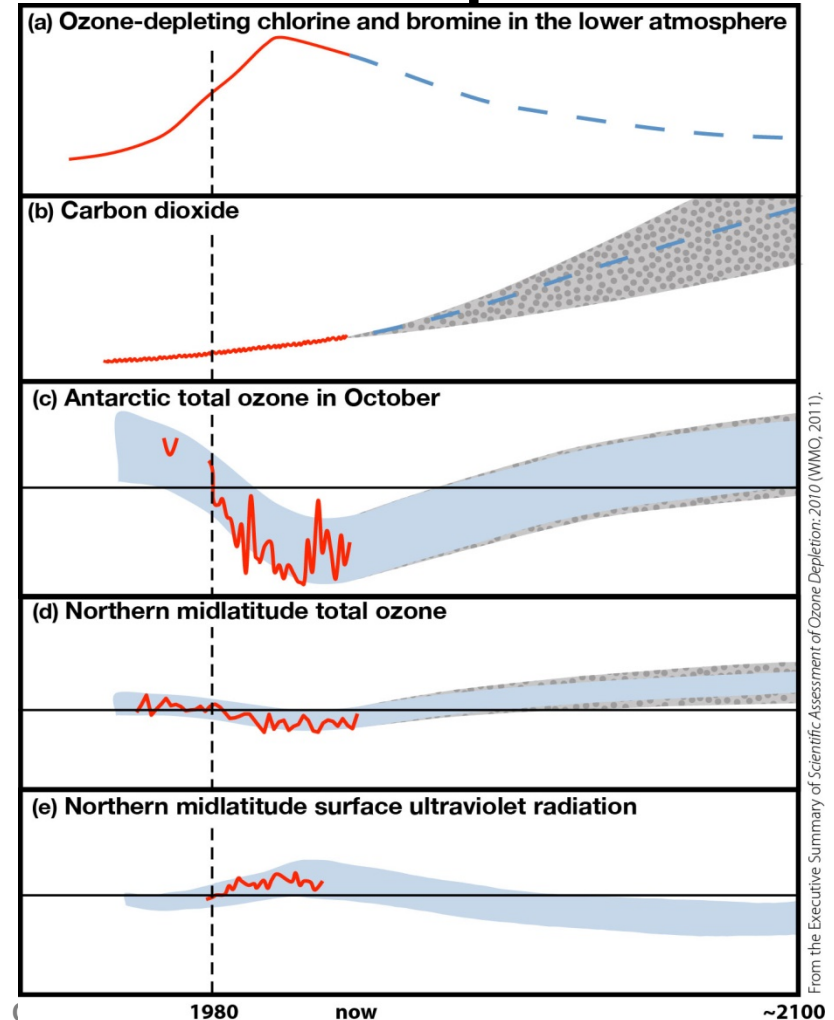


Depletion of Ozone Layer

Current Rate of Ozone Depletion

- Decrease in rate of ozone depletion (since 1997)
- Slowing of buildup of harmful Cl^- from CFCs
- Ozone hole is still growing, but...

Models anticipate restoration of “normal” balance of ozone in stratosphere by 2050



Acid Precipitation

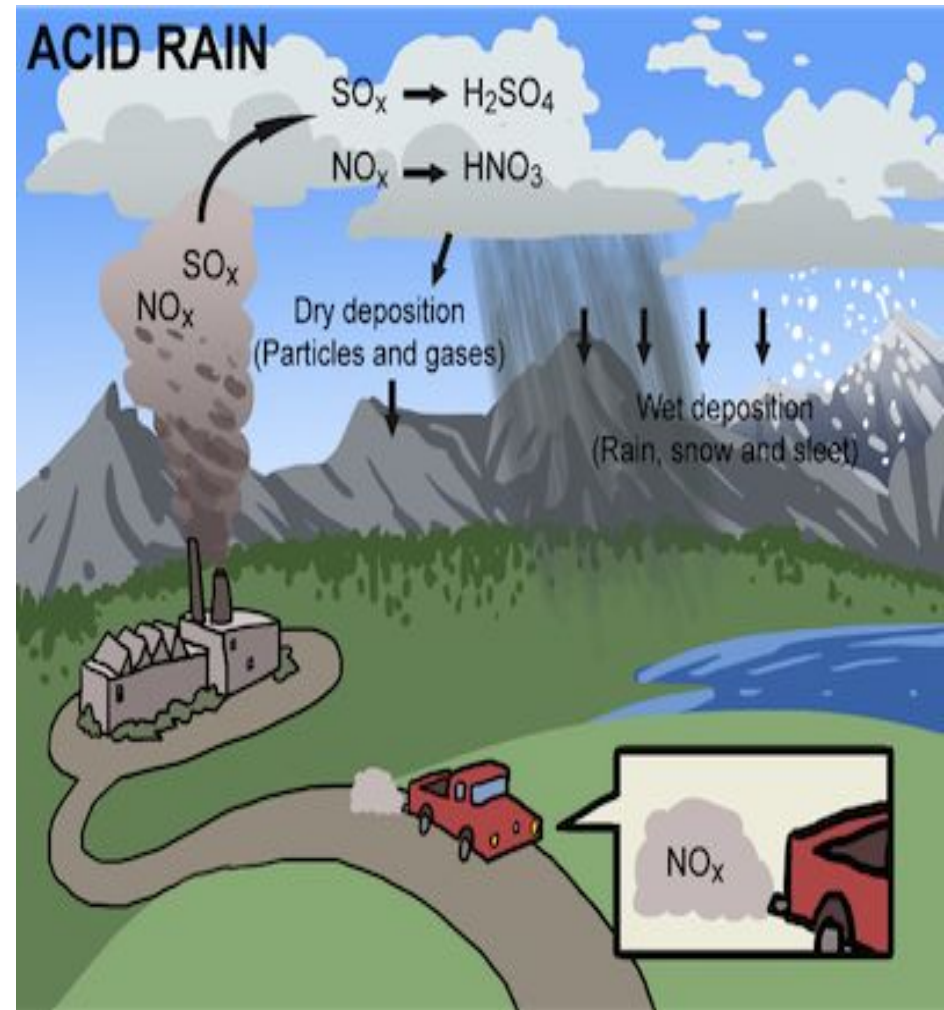




Acid Precipitation

Definition of Acid Rain

- Precipitation that has a pH of less than that of natural rainwater (which is about 5.6) due to dissolved oxides of sulphur, nitrogen and carbon.
- It is formed when oxides of sulphur, carbon and nitrogen, as gases or fine particles in the atmosphere, combine with water vapour and precipitate as sulphuric, carbonic or nitric acids in rain, snow, or fog.





Acid Precipitation

Causes of Acid Rain

- Natural Sources
 - Emissions from volcanoes , and biological processes that occur on the land, in wetlands, and from decomposing organic materials , in the oceans contribute acid-producing gases emission to the atmosphere
 - Effects of acidic deposits have been detected in glacial ice thousands of years in remote parts of the globe





Acid Precipitation

Causes of Acid Rain

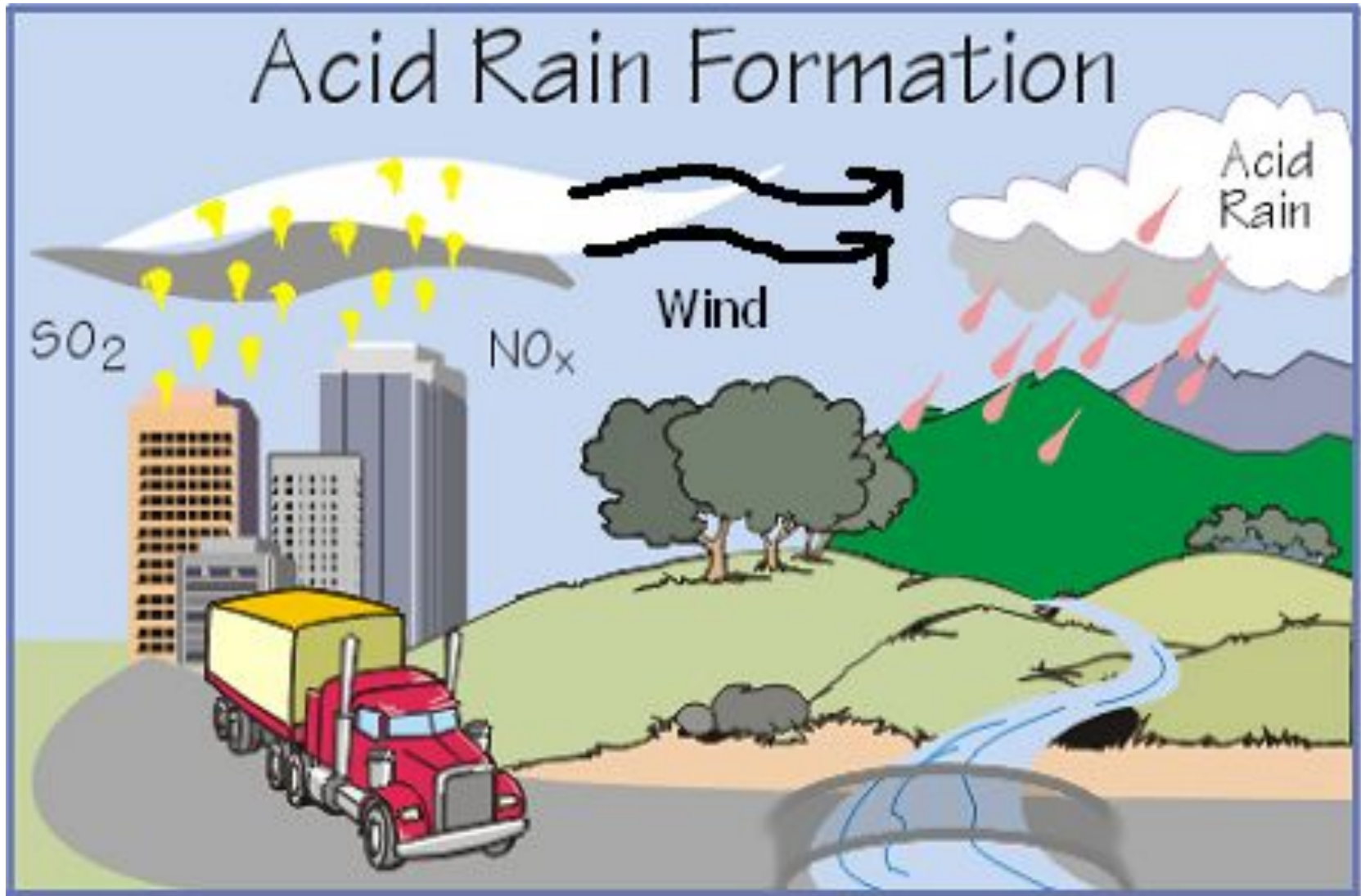
- The principal cause of acid rain is from **human sources**
 - Industrial factories, power-generating plants and vehicles
 - Sulphur dioxide and oxides of nitrogen are released during the fuel burning process (i.e. combustion)



MSN Encarta



Acid Precipitation





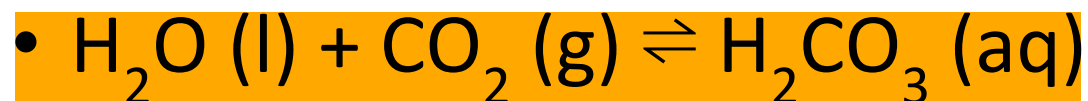
Acid Precipitation

Formation of Acid Rain

- When water vapour condenses, or as the rain falls, they dissolve in the water to form sulphuric acid (H_2SO_4) and nitric acid (HNO_3).
- in this way, it causes precipitation to become acidic, forming acid rain

Chemical reactions

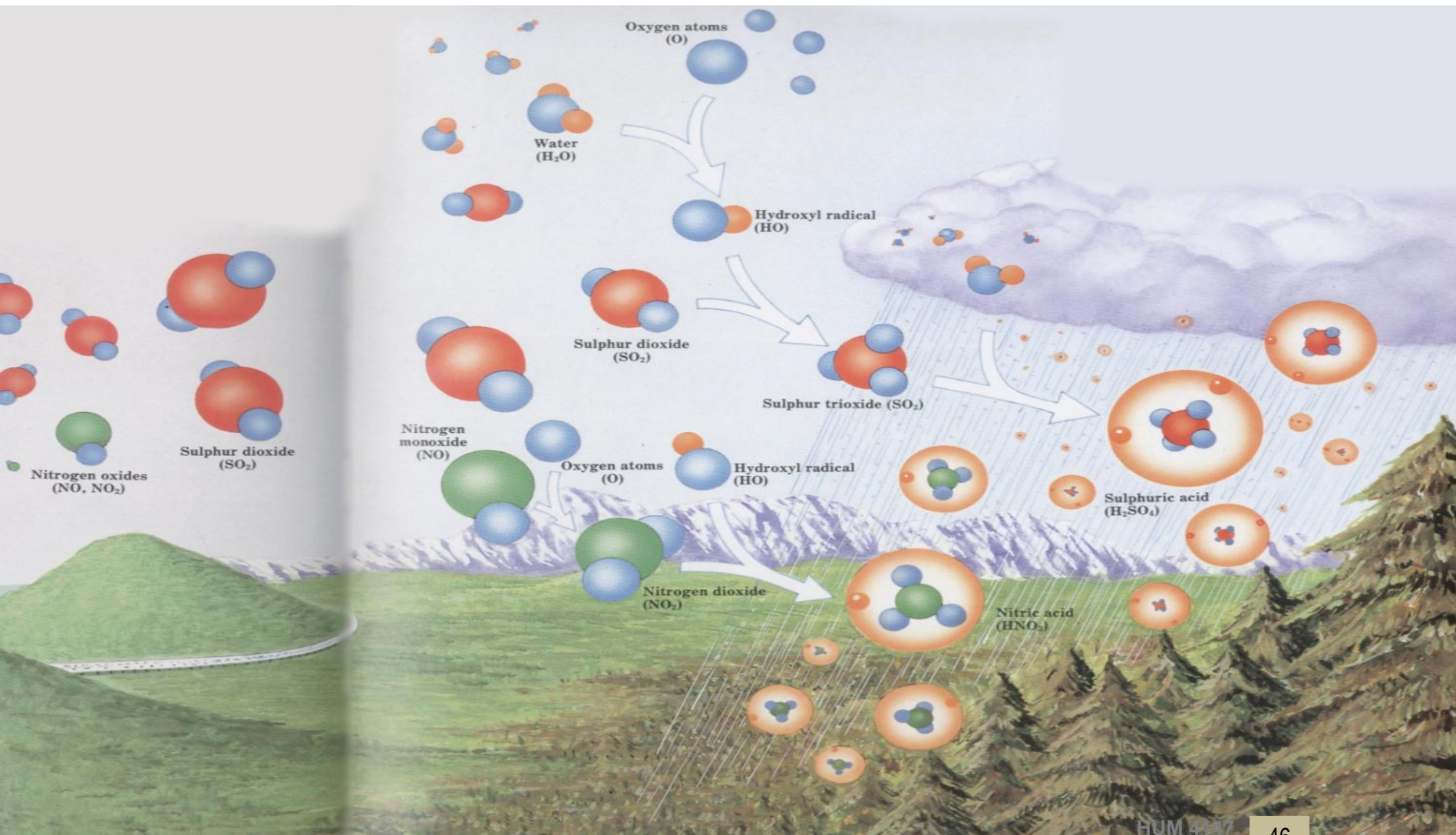
The following reaction will clarify the acid formation reaction:





Acid Precipitation

Formation of Acid Rain





Acid Precipitation: Effect

Affected Areas

- Industrial acid rain is a substantial problem in Northeastern part of the United States ,China, Eastern Europe and Russia and areas down-wind from them.
- Acid rain from power plants in the Midwest United States has also harmed the forests of upstate New York and New England.
- This shows that the effects of acid rain can spread over a large area, far from the source of the pollution



Acid Precipitation: Effect

Damage to Plant and animals

Damage to Chemical balance in soil

Damage to Aquatic Life

Damage to Heritage Life

Damage to Metals



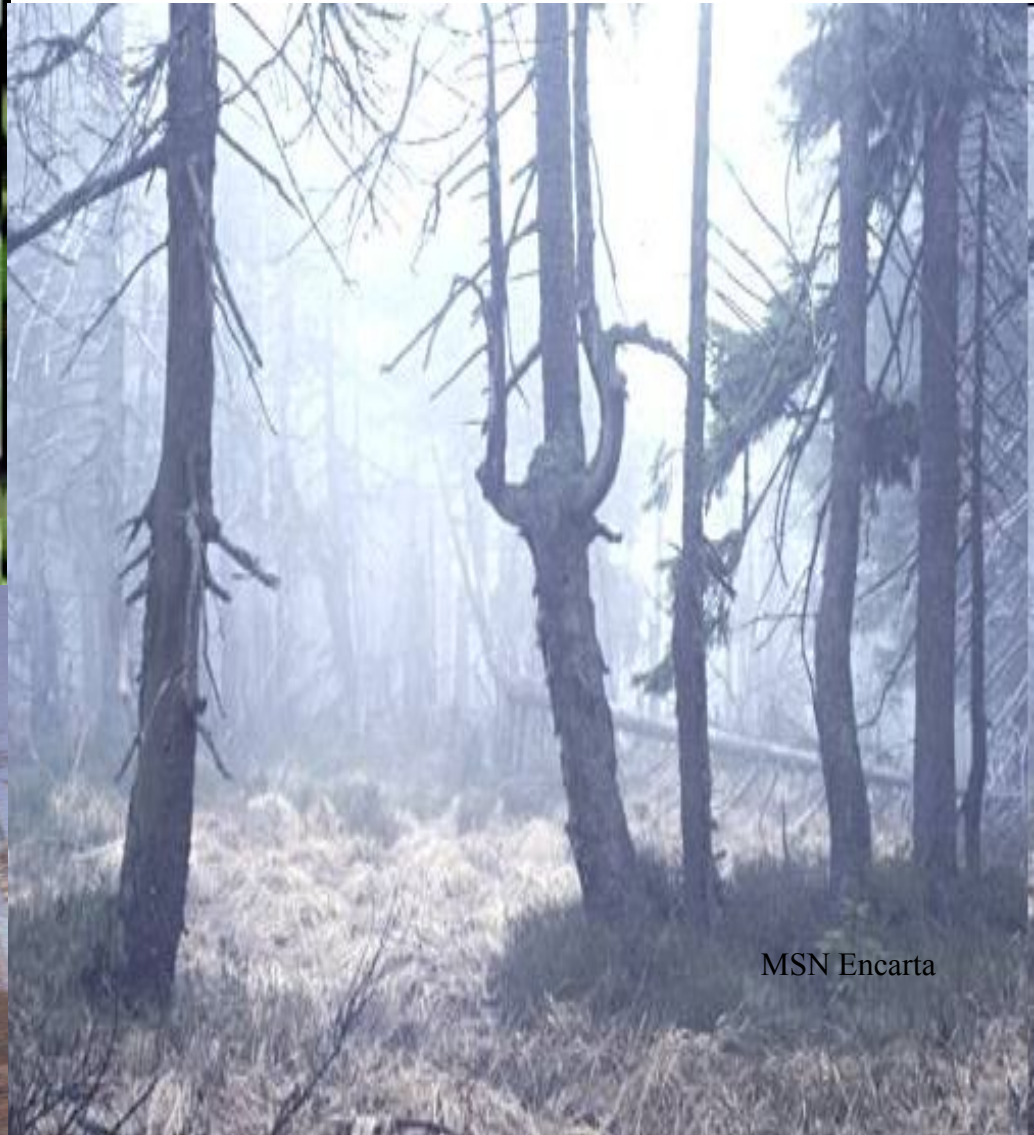


Acid Precipitation: Effect

- Harmful to vegetation
 - Increased acidity in soil
 - Leeches nutrients from soil, slowing plant growth
 - Leeches toxins from soil, poisoning plants
 - Creates brown spots on the leaves of trees, impeding photosynthesis
 - Allows organisms to infect through broken leaves



Acid Precipitation: **Effect**



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Acid Precipitation: Effect

- Deteriorating Buildings
- Accelerates weathering in metal and stone structures
 - E.g. Parthenon in Athens, Greece; Taj Mahal in Agra, India



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<http://www.lauraknauth.com/photos/france/thinker.jpg>



Acid Precipitation: Effect

- **Affects human health**
 - **Respiratory problems**, asthma, dry coughs, headaches and throat irritations.
 - Leeching of toxins from the soil by acid rain can be absorbed by plants and animals. When consumed, these **toxins affect humans severely, causing brain damage, kidney problems, and Alzheimer's disease which has been linked to people eating "toxic" from animals and plants.**



Acid Precipitation: Preventive Measures

- Reduce amount of sulphur dioxide and oxides of nitrogen released into the atmosphere
 - Use less energy (hence less fuel burnt)
 - Use cleaner fuels
 - Remove oxides of sulphur and oxides of nitrogen before releasing
 - Flue gas desulphurization
<https://www.youtube.com/watch?v=3HSc2uJ4cEs>
 - Catalytic Converters
https://www.youtube.com/watch?v=DAd_lrXiYCU
- <https://www.youtube.com/watch?v=IJiznlz5buc&t=72s>





Acid Precipitation: **Preventive Measures**

- **Flue Gas Desulphurization (FGD)**

Most FGD technologies use an alkali sorbent, such as limestone (calcium carbonate), quicklime (calcium oxide), hydrated lime (calcium hydroxide), or sometimes sodium and magnesium carbonate and ammonia, to capture the acidic sulfur compounds from the flue gas.

Removes sulphur dioxide from flue gas (waste gases)

Consists of a wet scrubber and a reaction tower equipped with a fan that extracts hot smoky stack gases from a power plant into the tower

Lime or limestone (calcium carbonate) in slurry form(semi-liquid) is injected into the tower to mix with the stack gases and reacts with the sulphur dioxide present.

Gas phase	Liquid phase	Solid phase	Rate determining steps
$\text{SO}_2 \rightarrow \rightarrow$	$\text{SO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{HSO}_3^-$ $\text{HSO}_3^- \rightleftharpoons \text{H}^+ + \text{SO}_3^{2-}$		Absorption of SO_2
$\text{O}_2 \rightarrow \rightarrow$	$\text{HSO}_3^- + 1/2\text{O}_2 \rightleftharpoons \text{H}^+ + \text{SO}_4^{2-}$ $\text{HSO}_4^- \rightleftharpoons \text{SO}_4^{2-} + \text{H}^+$		Oxidation of HSO_3^-
$\text{CO}_2 \rightarrow \rightarrow$	Ca^{2+} $\text{CO}_2 + \text{H}_2\text{O} \rightleftharpoons \text{HCO}_3^- + \text{H}^+$ $\text{HCO}_3^- \rightleftharpoons \text{CO}_3^{2-} + \text{H}^+$ $\text{H}_2\text{O} \rightleftharpoons \text{H}^+ + \text{OH}^-$	CaCO_3	Dissolution of limestone
	$\text{Ca}^{2+} + \text{SO}_4^{2-} + 2\text{H}_2\text{O} \rightarrow \rightarrow$	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$	Crystallization of gypsum

Catalytic converter

- A catalytic converter is an exhaust emission control device that converts toxic gases and pollutants in exhaust gas from an internal combustion engine into less-toxic pollutants by catalyzing a redox reaction.
- It filters out harmful byproducts in the exhaust gases and burns them up.
- It uses basic redox reactions to reduce the pollutants a car makes.
- It converts around 98% of the harmful fumes produced by a car engine into less harmful gases. It is composed of a metal housing with a ceramic honeycomb-like interior with insulating layers

- $\text{NO}_x \rightarrow \text{N}_x + \text{O}_x$
- $2\text{CO} + \text{O}_2 \rightarrow 2\text{CO}_2$
- $2\text{NO}_2 \rightarrow \text{N}_2 + 2\text{O}$
- $\text{C}_x\text{H}_{2x+2} + [(3x+1)/2] \text{O}_2 \rightarrow x\text{CO}_2 + (x+1) \text{H}_2\text{O}$
- **Platinum and palladium** take part in oxidation reactions. These reduce hydrocarbons (HC) and carbon monoxide (CO) in exhaust. First, carbon monoxide and oxygen combine to form carbon dioxide (CO₂). Then, unburned hydrocarbons and oxygen combine to form carbon dioxide and water.

Catalytic Converter Operation





Acid Precipitation: **Preventive Measures**

- Use cleaner fuels
 - Coal that contains less sulphur
 - "Washing" the coal to reduce sulphur content
 - Natural Gas





Acid Precipitation: **Preventive Measures**

- **Use other sources of electricity** (i.e. nuclear power, hydro-electricity, wind energy, geothermal energy, and solar energy)



http://upload.wikimedia.org/wikipedia/commons/4/4e/Nuclear_Power_Plant_Cattenom.jpg



Acid Precipitation: **Preventive Measures**

- **Liming**
 - Powdered limestone/limewater added to water and soil to neutralize acid
 - Used extensively in Norway and Sweden
 - Expensive, short-term remedy



**Thank you for your
attention**

Welcoming Q & A !!